Review notes · for you

Things that sense: unit check

IoT · Level 1 · Unit check

Materials
Paper and a pen. No device needed.

Learning goals

- Machines that notice
- The sensors in your phone
- Measure it right
- Connected things, safely

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Quiz sheet

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Name _____ Class _____ Date _____

1 **A greenhouse fan switches on when it gets too hot. Which part is the sensor?**

◯ The thermometer that reads 34 °C ◯ The small computer with its rule ◯ The motor that turns the fan

2 **Match each thing people can't sense to the sensor that notices it.**

The Earth's magnetism	• • • •	A parking sensor
Sound too high to hear	• • • •	A compass sensor
Infrared light bouncing off your ear	• • • •	A carbon monoxide alarm
A gas with no colour or smell	• • • •	A proximity sensor

3 **A phone camera is one big light sensor that stores a tiny picture.**

◯ True ◯ False

4 **Read the scale.**

Between the marks 0 and 50 there are 10 gaps, so each gap is worth _____. A reading 7 gaps above 0 is _____.

Word bank: 7 · 10 · 5 · 35

5 **Which of these make a home Wi-Fi camera safer?**

☐ Sharing its link in a big group chat ☐ Changing the factory password ☐ Turning on updates
☐ Switching off features you don't use ☐ Keeping the password on a sticker on the camera

6 **Put an automatic hand dryer's steps in order.**

☐ Its sensor notices them
☐ Your hands move under the dryer
☐ The fan blows warm air
☐ You pull your hands away, and it stops
☐ The small computer checks its rule

7 **Five classroom temperatures: 28, 29, 85, 28 and 30 °C. What should you report?**

◯ 40 °C, the average of all five readings ◯ About 29 °C, saying the 85 was left out and why
◯ 85 °C: the highest reading is the safest to report

8 **If you tap Don't allow when an app asks for your microphone, you can change your mind later in Settings.**

◯ True ◯ False

9 **You're on a farm with no mobile signal, but you saved the map before you left. Which sensor can still show where you are?**

◯ The compass sensor ◯ The proximity sensor ◯ The GPS receiver

10 **Kofi's soil log shows 0 % between two readings of 50 %, an hour apart. Which are good next steps?**

☐ Check the sensor's wires ☐ Water the garden at once ☐ Measure again, or compare with another tool
☐ Mark it, and note why it was left out ☐ Delete it quietly

Check your answers

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1 **The thermometer that reads 34 °C**

The thermometer notices the heat and gives a number: that's sensing.

2 **The Earth's magnetism — A compass sensor**
Sound too high to hear — A parking sensor
Infrared light bouncing off your ear — A proximity sensor
A gas with no colour or smell — A carbon monoxide alarm

Machines can notice what our senses miss: the compass feels magnetism, the parking sensor times echoes of very high sound, the proximity sensor watches its infrared light bounce back, and the alarm detects carbon monoxide.

3 **False**

It's millions of tiny light sensors in a grid, and the phone stores every pixel as about three numbers: red, green and blue.

4 **Between the marks 0 and 50 there are 10 gaps, so each gap is worth 5. A reading 7 gaps above 0 is 35.**

50 ÷ 10 = 5 a gap, and 7 × 5 = 35. Answering 7 counts each gap as 1; 10 is the number of gaps, not what each is worth.

5 **Changing the factory password; Turning on updates; Switching off features you don't use**

A new password beats the attackers' lists, updates fix known weak spots, and fewer features mean fewer ways in.

6 **Top to bottom: 2, 1, 4, 5, 3**

Sense, decide, act: the sensor notices first, the computer decides, and the fan acts.

7 **About 29 °C, saying the 85 was left out and why**

No classroom reaches 85 °C, so it's a glitch, and it drags the average up to 40.

8 **True**

Every permission can be turned on or off later in your phone's Settings, so saying no first is safe: you can always allow it if the app really needs it.

9 **The GPS receiver**

GPS listens to satellites and needs no mobile signal or data, and the map was already saved.

10 **Check the sensor's wires; Measure again, or compare with another tool; Mark it, and note why it was left out**

Soil can't dry out and get wet again in an hour, so check the sensor, measure again and mark the glitch openly.